Explainable Predictive Analytics for Employee Attrition Risk Assessment

An Explainable AI (XAI) & HR Decision-Support System Powered by ML and SHAP

PROJECT DOCUMENTATION

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1. Introduction

Employee turnover creates direct recruitment expense and indirect operational loss through disrupted teams, delayed delivery and the departure of experienced staff. A predictive model can help HR identify elevated risk, but a probability alone does not explain the conditions behind the prediction. This project combines predictive analytics with explainable AI so that each score can be reviewed in the context of its strongest contributing factors.

1.1 Problem Statement

Conventional attrition models often operate as black boxes. They may classify an employee as high risk without showing why the model reached that result. This limits trust and can encourage inappropriate or punitive use. The project addresses this gap through SHAP explanations, human review and counterfactual simulation.

1.2 Project Objectives

- Train and compare multiple classification models on a consistent held-out test set.
- Select a model that balances predictive performance, stability and explainability.
- Provide global and employee-level SHAP explanations.
- Deliver an interactive full-stack application for risk review, scenario testing and retention-action tracking.
- Preserve human oversight and prevent the model from becoming an automated employment decision maker.

1.3 Scope

The application is an academic decision-support prototype based on the IBM HR Analytics benchmark. It evaluates risk patterns in the supplied data and demonstrates how transparent predictions can support structured retention conversations. It does not establish causation or validate performance on a live organization.

2. End-to-End Architecture

The solution follows a five-layer analytical workflow from dataset intake to interactive decision support.

1. Data Layer

IBM HR benchmark dataset with 1,470 records and 35 employee attributes.

2. Preprocessing

One-hot encoding for categorical variables, StandardScaler normalization and an 80/20 stratified split.

3. ML Modeling

Comparable evaluation of Logistic Regression, Decision Tree, Random Forest and Gradient Boosting.

4. Explainability

SHAP TreeExplainer for global feature rankings and individual feature attributions.

5. Application

React and Vite interface connected to FastAPI endpoints, with SQLAlchemy and SQLite persistence.

2.1 Technology Stack

Layer	Technology	Purpose
Frontend	React, Vite, Tailwind CSS	Interactive HR risk dashboard
API	Python FastAPI	Asynchronous REST services
ML/XAI	scikit-learn, SHAP	Prediction and explanation
Persistence	SQLAlchemy, SQLite	Employees, predictions and actions

3. Dataset and Preprocessing

The project uses the IBM HR Analytics Employee Attrition and Performance benchmark. The dataset contains 1,470 records across 35 attributes covering employee profile, role, compensation, tenure, satisfaction and work patterns.

3.1 Preparation Workflow

- Categorical fields are converted through one-hot encoding.
- Numeric fields are normalized with StandardScaler where required by the preprocessing pipeline.
- The data is divided using an 80/20 stratified partition so the class distribution remains consistent.
- The held-out test set contains 294 records and supports like-for-like model comparison.

3.2 Model Candidates

The benchmark compares a linear classifier, a single decision tree, a bagging ensemble and a boosting ensemble. This mix provides interpretable baselines and nonlinear models that capture interactions among employee attributes.

4. Model Evaluation

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	75.51%	35.63%	65.96%	46.27%	80.35%
Decision Tree	77.21%	35.71%	53.19%	42.74%	65.17%
Random Forest (Selected)	**82.65%**	**45.24%**	**40.43%**	**42.70%**	**78.55%**
Gradient Boosting	85.03%	57.89%	23.40%	33.33%	81.46%

Selection rationale. Random Forest is used by the application because it combines stable ensemble behavior with direct SHAP TreeExplainer support. Gradient Boosting records higher accuracy and ROC-AUC in the stored benchmark, but its lower recall and F1-score show that the choice should not be based on accuracy alone.

5. SHAP Feature Attribution

SHAP assigns each feature a contribution relative to the model's baseline prediction. Positive values push the predicted attrition probability upward, while negative values reduce it. The explanation remains a description of model behavior, not proof that a feature causes resignation.

5.1 Global Importance

Feature	SHAP importance	Interpretation
OverTime = Yes	0.0532	Strongest overall risk-increasing signal
StockOptionLevel	0.0372	Lower option tiers can increase predicted risk
Age	0.0330	Younger cohorts show higher modeled volatility
MonthlyIncome	0.0313	Compensation level contributes to flight-risk estimates
YearsWithCurrManager	0.0272	Longer manager relationships can act as a protective signal

5.2 Local Explanations

For an individual employee, the application displays the strongest risk-increasing and risk-reducing factors. The reviewer can compare these attributions with the employee's current context before considering any supportive action.

5.3 Interpretation Rules

- Use an explanation to understand a model score, not to label an employee.
- Do not treat a high SHAP value as causal evidence.
- Confirm relevant context through respectful human dialogue.
- Avoid using protected personal characteristics or explanations for punitive action.

6. Production Implementation

6.1 Executive Risk Radar

The dashboard presents workforce-level indicators, including total evaluated records, the 16.1% historical attrition benchmark, predicted-risk distribution and department or role summaries.

6.2 Workforce Directory

Users can search and filter employees by role, overtime status and calibrated risk tier. The application uses Low (<30%), Medium (30-59%) and High (>=60%) risk categories to organize review queues.

6.3 Employee Detail and Explanation

The employee view combines the predicted probability, risk category and local SHAP drivers. This design keeps the score and its explanation together so reviewers do not act on a probability in isolation.

6.4 Model Benchmark

The benchmark module exposes the measured test-set results for all four models, including accuracy, precision, recall, F1-score, ROC-AUC and confusion-matrix values stored with the trained artifacts.

6.5 Retention Action Tracker

Reviewers can record supportive retention activities such as workload balancing, compensation review and manager check-ins. Actions remain subject to human judgment and follow-up.

7. What-If Simulation Engine

The simulator creates a counterfactual copy of an employee record, applies selected changes and recalculates the risk score with the trained Random Forest model. The original record remains the comparison point.

7.1 Supported Adjustments

- Overtime status and monthly income.
- Work-life balance, job satisfaction and environment satisfaction.
- Stock-option level, distance from home and job involvement.
- Salary-hike percentage and years in the current role.

7.2 Simulation Output

The API returns the original and simulated risk scores, their categories, the modeled risk change, the applied adjustments and the top SHAP drivers for the simulated case. A lower simulated probability indicates a modeled change under the chosen assumptions; it does not guarantee that the intervention will prevent resignation.

7.3 Operational Workflow

1. Select an employee or cohort for review.
2. Inspect the score, local SHAP factors and relevant work context.
3. Test a realistic supportive intervention in the simulator.
4. Discuss the option with appropriate human oversight and employee consent.
5. Record an agreed action and schedule a follow-up review.

8. Responsible AI and Ethical Governance

- Attribution is not causation. Predictions should guide supportive review, not automatic employment action.
- The model must not automate firing, promotion denial or disciplinary decisions.
- Reviewers should minimize data collection and use job-relevant attributes only.
- Retention activities require informed human judgment, empathy and appropriate consent.
- Access to employee-level predictions and explanations should be restricted and audited.

9. Validation and Known Limits

- The evaluation uses a stratified 20% holdout of the IBM benchmark, not a live enterprise deployment.
- Model metrics describe this dataset and split; they do not guarantee the same performance in another organization.
- Class imbalance means accuracy alone can overstate practical performance. Precision, recall and F1-score remain important.
- SHAP explains the fitted model's behavior but cannot prove a causal relationship.
- Counterfactual results are model-based estimates and require real-world follow-up before policy conclusions.

10. Conclusion

The project demonstrates how predictive analytics can support HR review without hiding the model's reasoning. Random Forest provides the deployed risk estimate, SHAP connects each score to interpretable feature contributions, and the React-FastAPI application turns those outputs into a reviewable workflow. The strongest value comes from combining transparent evidence with careful human judgment and supportive retention practices.

11. Running the Project

11.1 Backend

```
cd backend
python -m uvicorn api.main:app --host 0.0.0.0 --port 8000 --reload
```

FastAPI documentation is available at http://localhost:8000/docs.

11.2 Frontend

```
cd frontend
npm run dev
```

The Vite development interface is available at http://localhost:5173.

11.3 Main User Flow

- Open the executive dashboard and review aggregate risk indicators.
- Filter the workforce directory by risk tier, role or overtime status.
- Open an employee record and inspect the local SHAP explanation.
- Test a supportive change in the what-if simulator.
- Record any approved retention action and its planned follow-up.

12. Project Team

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